
To Forge or Not to Forge: Predicting Task-Level Fine-Tuning Gains from Ten-Example Pilots

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Abstract

1 Fine-tuning a small specialist is the default path to cheap task-specific capability,
2 yet the upstream decision — is this task worth fine-tuning at all? — is still made by
3 intuition or by paying for the full run. Under one frozen protocol (Qwen2.5-1.5B-
4 Instruct, fixed LoRA recipe and budgets, greedy decoding, execution-based veri-
5 fiers) we pair a ten-example pilot with a full fine-tune on 61 tasks — 11 anchored
6 in the small-model post-training literature, 50 drawn by a pre-registered filter from
7 Super-NaturalInstructions — and take the measured gain as ground truth. Fine-
8 tuning helps on 53 of 61 tasks (87%), but 23 tasks start at $\text{pass}@1 = 0$, where the
9 gain is format unlock rather than new capability. The pilot *ranks* gain magnitude
10 (Spearman 0.755, CI [0.60, 0.86]), and the ranking has budget value: spending
11 $K=20$ full fine-tunes in pilot order captures 0.72 of all positive-gain mass, against
12 0.33 random and 0.74 oracle. The pre-registered binary go/no-go endpoint is pub-
13 lished as a negative (leave-one-task-out AUC 0.755, CI [0.50, 0.93]). A declared
14 revision locates the mechanism: removing the one post-decision feature does *not*
15 rescue it (0.663, CI [0.43, 0.84]), while a three-feature pre-decision model does
16 (0.767, CI [0.65, 0.88], permutation $p=0.004$) — the registered negative was an
17 over-parameterised estimator, not an absence of signal. Binary framing was in any
18 case the wrong target: a *perfect* gate is worth 2.4% over always forging, and the
19 probe is not free (median 0.17 of full training-plus-eval cost, 0.53 including the
20 base evaluation it needs). Instruction embeddings predict at chance (0.512): you
21 have to run the probe. We characterise the no-go tail, including negative transfer
22 on heavily post-trained bases (GSM8K -0.23 despite in-domain training), and
23 release the benchmark, protocol, analysis code, and all sealed artifacts. Every
24 measurement here was produced by an autonomous discover-pilot-decide-train-
25 validate system; without it the 61-task ground-truth table would not exist.

26 **Keywords:** fine-tuning, evaluation, small language models

27 1 Autonomous research result

28 1.1 Hypothesis

29 Under a single frozen LoRA recipe on a 1.5B instruction-tuned base, can a ten-example probe decide
30 *which* tasks to fully fine-tune, and is that decision a binary gate or a ranking for a budget of K slots?
31 We pre-register a binary label $\text{go} \Leftrightarrow \Delta_{\text{full}} > 0$ and a secondary ranking endpoint $\rho(\Delta_{\text{pilot}}, \Delta_{\text{full}})$,
32 with the honesty clause that a non-significant AUC is published as a negative. A later, declared
33 revision (v2) asks the same questions after removing post-decision features from the classifier, and
34 quantifies the *utility* of a perfect gate versus always forging. A correction to that revision (the

addendum) separates the effect of removing leakage from the effect of one declared bug fix; we report both.

1.2 Method

Protocol and sample. Base model Qwen2.5-1.5B-Instruct [Qwen Team, 2024], weights pinned at birth; LoRA rank 16, $\alpha = 32$; sequence 4096; effective batch 16; pilot 10×100 ; full cap 8000 / 3 epochs; eval 200, greedy. pass@8 ($k=8$) is a signal (1600 generations per task), never the label. The sealed sample is $n=61$ (11 literature-layer + 50 SuperNI, seed 20260822): 53 registered go / 8 no-go (87%). BIRD and APPS are out, excluded for logistics, before any label existed. All numbers below that are not marked “registered” come from ANALYSIS_RESULTS_v2 and its correction file ANALYSIS_RESULTS_v2_ADDENDUM (CPU revision; the addendum supersedes the v2 attribution wherever they differ). Registered AUC/CI remain the v1 sealed row.

Predictors. The *new main* classifier is leave-one-task-out logistic regression on the pre-decision vector only: Δ_{pilot} , base pass@1, base pass@8, headroom, pilot-loss descriptors, format-compliance of the *base* split, and train_n. gen_len.full.median is reported as a post-hoc oracle upper bound and is *not* used for decisions. The registered 12-feature vector (which included that oracle and full_n) is kept as the registered row. Intervals are task-level bootstrap 95% CIs (seed 20260820, $B=1000$). A label-permutation test re-runs full LOTO $N=1000$ times (seed 20260826) and is reported as $(1 + \#\{\rho_{\text{perm}} \geq \rho_{\text{obs}}\}) / (1 + N)$.

One declared bug fix, found after the revision contract. The revision code also changed one feature *definition*: the v1 format-compliance scalar $1 - \text{unparseable}/n$ silently returns 1.0 for any task whose compliance block carries no unparseable key. Exactly two of the 61 tasks are like that — GSM8K, which v1 special-cased, and MATH, which it missed. The revision adds the matching MATH branch (boxed/ $n = 0.55$). This is a bug fix, not a task-specific choice, but it was found *after* ANALYSIS_PREREG_V2 was written, it is not one of that page’s declared items, and it lands on one of only two substantive negatives. We therefore report every affected number under both definitions and never fold its effect into “removing leakage” (Table 1). We also report L2-CV LOTO, a 3-feature model (Δ_{pilot} , base_pass1, headroom), and a no-pass@8 feature set. Instruction embeddings (MiniLM, CPU) plus SuperNI Categories/Source are the semantic control; the original four scalars are the metadata control. McNemar exact tests on paired eval jsonl induce a three-way label (go / no-go / undetermined) as a revision-period analysis; the registered label stays $\Delta_{\text{full}} > 0$. Sensitivity (b) (drop EM-floor rows) is computed but *not* used as supporting evidence (wide CI).

Related work (lite). Training-free transferability estimators (LogME, LEEP, TransRate) score a frozen representation against target labels without fine-tuning [You et al., 2021, Nguyen et al., 2020, Huang et al., 2021]. We did *not* run them: they select among source models, whereas we select among *tasks* given one frozen recipe. That is a limitation and a listed future comparison. Multifidelity methods (successive halving, Hyperband, learning-curve extrapolation) [Jamieson and Talwalkar, 2016, Li et al., 2018] treat cheap runs as a fidelity axis. Our probe is a *single* fidelity point; a fidelity scan is future work.

1.3 Evidence

Finding 1: forging is the default in this regime. 53 of 61 tasks (87%) have $\Delta_{\text{full}} > 0$. This is Qwen2.5-1.5B-Instruct, one LoRA recipe, greedy pass@1, gate $\Delta_{\text{full}} > 0$ — not a claim about 7B models or full FT. 23 of 61 tasks have base_pass1 = 0; most of those gains are format unlock (the probe teaches the extractor), not new reasoning.

Finding 2: the probe ranks gain, and the ranking has budget value. Spearman $\rho(\Delta_{\text{pilot}}, \Delta_{\text{full}}) = 0.755$, CI [0.595, 0.864] (registered secondary; excludes 0). Partial ρ controlling for base_pass1 is 0.832 (residualise the raw deltas on base, then rank; the fully rank-based partial formula gives 0.777 — either way the ranking is not a restatement of base accuracy). Figure 1 is the operational plot. Ranking by Δ_{pilot} and spending a budget of K full fine-tunes captures 0.42 / 0.72 / 0.83 of all positive-gain mass at $K=10, 20, 30$, against random 0.16 / 0.33 / 0.49 and oracle 0.45 / 0.74 / 0.90. At $K=20$ the probe is within three points of oracle. 38% of tasks have

85 $|\Delta_{\text{pilot}}| \leq 0.02$ (a dumb probe); the ranking still works because the mass sits in the tails. Figure 2
 86 shows the same pairing; GSM8K/MATH sit in Q3 and the EM-floor quartet sits at the origin.

87 **Finding 3: the binary gate, three ways.** *Registered row.* LOTO on the original 12-feature vec-
 88 tor (including post-decision `gen_len.full.median`) has AUC 0.755, CI [0.500, 0.929], covering 0.5
 89 — a negative under the honesty clause. (The point 0.7547 and Spearman 0.7553 both print as
 90 0.755; that is a rounding coincidence, not one statistic.) *Revision rows, decomposed.* Dropping
 91 the post-decision features *lowers* the AUC to 0.663, CI [0.433, 0.839], permutation $p=0.057$ — still
 92 a negative. The jump to 0.844 [0.733, 0.943], $p=0.002$, comes from the declared MATH format-
 93 compliance fix above: one cell of the 61×10 design matrix, worth +0.182 AUC on its own. The
 94 leaked feature was *helping* the registered row, so the v1 negative cannot be read as “leakage.” It
 95 is read correctly as *over-parameterisation*: 12 features against 8 negatives. The parsimonious pre-
 96 decision model — three features, no format-compliance term, and therefore invariant to both the
 97 leak and the fix — is significant at 0.767 [0.646, 0.879], $p=0.004$. That row, not 0.844, is the defen-
 98 sible pre-decision estimate. L2-CV (0.875) and the no-pass@8 set (0.837) inherit the same +0.15
 99 dependence on the fix and are reported with their unfixed values in Table 1. None of these rows is
 100 stable at the second decimal: dropping a single substantive negative moves the 0.844 model to 0.685
 101 (MATH) or 0.714 (GSM8K), across a jackknife range [0.685, 0.871]. *Utility.* A perfect gate’s upside
 102 versus always-forging is $|\sum \text{negative } \Delta| / \sum \text{positive } \Delta = 0.024$. Net always-forge is +16.6 pass
 103 points; oracle-forge is +17.0. McNemar three-way labels are 45 go / 2 no-go / 14 undetermined
 104 ($\alpha=0.05$, uncorrected). With two significant negatives the determined-only AUC is not a serious
 105 gate estimate. Figure 3 is the pre-decision ROC.

106 **Finding 4: you still have to run the probe.** Instruction-sentence embeddings plus SuperNI cate-
 107 gory/source have AUC 0.512 [0.255, 0.746] — chance. The old four-scalar “text-only” AUC 0.309
 108 is a *metadata* control whose `eval_n` is constantly 200; it is not evidence that description features run
 109 below chance in any semantic sense. Single-feature `steps_to_0.01` is 0.748 [0.585, 0.881] (does not
 110 cover 0.5; five tests, *not* multiplicity-corrected). `gen_len.full.median` is 0.818 [0.528, 0.978] and is
 111 post-hoc.

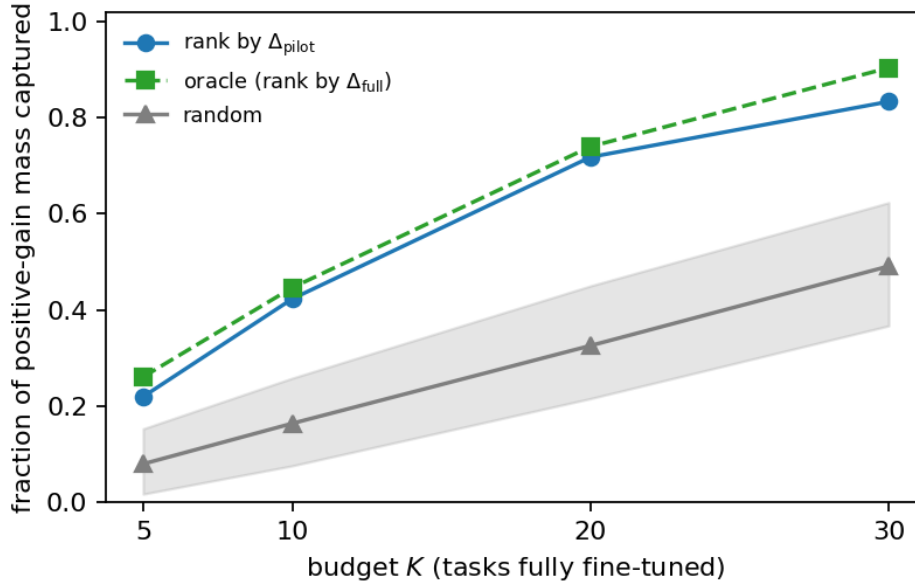


Figure 1: Budget allocation. K full fine-tunes, fraction of positive Δ_{full} mass captured.

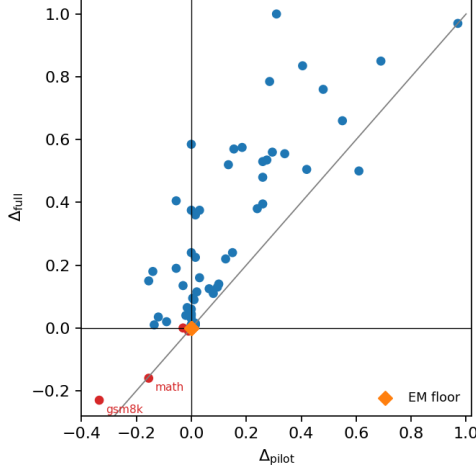


Figure 2: Pilot vs. full, cropped. Diamonds: EM-floor rows.

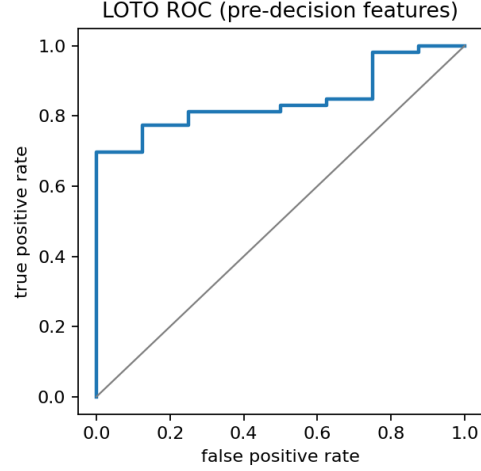


Figure 3: LOTO ROC, pre-decision features.

Table 1: Classifiers on the registered go label. “fix” is the declared MATH format-compliance branch; rows that contain no compliance term are invariant to it. The 3-feature row is the only significant pre-decision model that depends on neither the leak nor the fix.

Model	AUC (with fix)	95% CI	no fix	covers 0.5
Registered 12-feat (v1, leaked)	0.755	[0.500, 0.929]	0.755	yes
Pre-decision 10-feat	0.844	[0.733, 0.943]	0.663	no
L2-CV pre-decision	0.875	[0.763, 0.960]	0.722	no
3-feat (invariant)	0.767	[0.646, 0.879]	0.767	no
No pass@8	0.837	[0.721, 0.930]	0.675	no
Semantic (MiniLM)	0.512	[0.255, 0.746]	0.512	yes
Metadata (4 scalars)	0.309	[0.051, 0.675]	0.309	yes

Median wall $(S2+S3)/(S4+S5) = 0.17$ (IQR [0.11, 0.36]); including S1, 0.53 (IQR [0.35, 1.19]).
pass@8 is 1600 generations per task and is not required for the gate (no-pass@8 row).

1.4 Primary Result

In this regime the probe is a *ranker with measured budget value*, not a stoplight worth building operations around. Spending $K=20$ full slots by Δ_{pilot} harvests 72% of all positive gain, near the oracle 74%. The registered binary endpoint is negative and stays negative after the post-decision feature is removed (0.663); only a parsimonious three-feature pre-decision model is significant (0.767). Even that is almost useless as a *decision*, because a perfect gate saves 2.4% of positive-gain mass relative to always forging. The 87% base rate, 23/61 zero-base format unlocks, and McNemar’s two significant negatives are the same fact: no-go is scarce. Instruction text does not replace the probe (semantic AUC 0.51). We do not reopen BIRD/APPS, do not move the registered go threshold, and do not treat sensitivity (b) as evidence.

124 2 System Design and Meta-Analysis

125 2.1 Agent and Harness

126 We ran the study with two coupled pieces: a frozen Python harness that executes one task from
 127 package load through a sealed `metrics.json`, and an LLM coding agent that packed those tasks,
 128 drove the remote GPU worker, diagnosed failures, ran the pre-registered analysis, and drafted this
 129 manuscript. The experimental model inside the harness is Qwen2.5-1.5B-Instruct at a pinned re-
 130 vision; the coding agent’s weights are omitted here for double-blind review and are named in the
 131 disclosure for the chairs.

132 The harness is a single-task state machine S0–S6: load and hash-check the package; greedy-evaluate
 133 the unadapted base; train a 10-example LoRA probe; evaluate the probe; train the full LoRA; eval-
 134 uate the full adapter; seal metrics (systems block included). Resume is journal-boundary only: a
 135 killed full-training stage restarts at S4 with S0–S3 kept. The agent never edits the protocol file. It
 136 may add scripts, isolate a failed run directory, or stop and write a report.

137 Figure 4 is the loop that produced the sealed $n=61$ table: a task factory writes frozen packages;
 138 a serial queue feeds the S0–S6 runner; gates mark PARTIAL or `over_budget` without inventing
 139 labels; incremental then full harvest brings artifacts back; a pre-registered script seals the analysis.

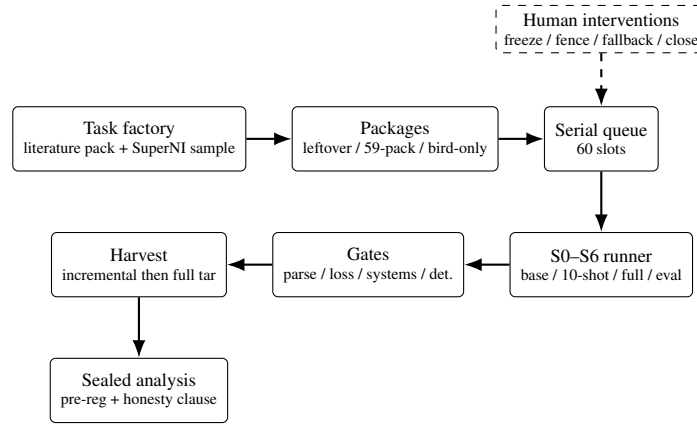


Figure 4: Factory → frozen packages → serial queue → S0–S6 runner → gates → harvest → sealed analysis. Dashed box: human intervention points (freeze, fence, fallback, close).

140 2.2 Tools

141 Compute was one 48GB-class GPU worker and one serial queue. The worker ran the harness
 142 under a terminal multiplexer; the agent supervised it with scheduled probes of `journal.jsonl`,
 143 `STATUS`, and `metrics.json`, because a remote job does not wake the session when it ends. Lo-
 144 cal tools were git on a single mainline (small commits, no force-push), an append-only ledger,
 145 pytest, and SHA-256 manifests. Execution-based verifiers (GSM8K, WinoGrande, Spider, later
 146 MATH/MBPP/DROP/TyDiQA/ARC/HellaSwag/PIQA and a SuperNI exact-match family) expose
 147 a uniform `verify(output, reference) -> {pass, parsed, note}`; unparseable is parsed
 148 is `None`, never a silent wrong. Each verifier has mutation tests: gold must pass, a known-wrong
 149 answer must fail, format variants of a correct answer must pass, garbage must unparse. GSM8K
 150 accepts ##### 1,000 as 1000 and rejects a hash without a number; WinoGrande will not treat the
 151 bare English article “a” as choice A; Spider compares execution multisets and treats a 30s timeout
 152 as fail with the exception type in note. The factory pins source revisions at pack time (dataset
 153 SHAs in the package `task.json`) and writes a per-package `MANIFEST.sha256`. SuperNI clones
 154 were frozen at 55a36563... before ids were committed; English+size filter left 890 surviving tasks,
 155 then the seed-20260822 draw of 50 with a `Source[0]` cap of 3. Code execution is a subprocess
 156 with a 10s cap, no network, and a 2 GiB virtual-memory limit.

157 2.3 Research Loop

158 The loop was not an open-ended search over architectures. It was discover (pack a frozen task), pilot
159 (10-shot LoRA), decide (record Δ_{pilot} as a signal, not a gate), train (full LoRA), validate (greedy
160 pass@1), then stop. The scientific hypothesis — whether that probe classifies go $\Leftrightarrow \Delta_{\text{full}} > 0$ —
161 was written down before the remaining production runs.

162 Protocol freeze came first. After three smoke runs the operator stopped a proposed `protocol_v3`:
163 the pilot 10×100 reaching loss $\sim 10^{-5}$ is recitation, and recitation is the probe, not a bug.
164 `protocol_v2.yaml` was never edited after that freeze; any later exception is a run-level flag, not a
165 silent yaml change.

166 Sample-frame correction came next. An earlier count treated 46 retestable *pool rows* as 46 tasks.
167 Deduped literature-layer tasks are 13 (3 already sealed, 10 new) plus 50 SuperNI tasks drawn by a
168 pre-registered English-and-size filter (seed 20260822, `Source[0] ≤ 3`). Intended $n=63$. A second
169 GPU worker was cancelled; the 60 new slots ran as one serial list. BIRD was packed later as its own
170 package; the 59-task package did not wait. Mini-Dev was never used.

171 The serial queue started 2026-08-22T09:34Z and drained 60/60 at 2026-08-24T00:22Z on that one
172 GPU (39 h wall). Of those 60 slots, 55 sealed `STATUS=ok` on the 59-task package; the rest were PAR-
173 TIAL or `over_budget` (TyDiQA and task419 S4 OOM, ARC-Easy unparseable at 5% with metrics,
174 BIRD S0 field error, APPS 3 h S1-only). Three earlier smoke runs (GSM8K, WinoGrande, Spider)
175 already counted as production data under the freeze. A per-task 3 h wall marks `over_budget`; gates
176 run after S6 and do not block the next task. Determinism uses a 200-row rerun on list items 1 and
177 every 10th, otherwise 30 rows. Incremental harvest pulled a 10-directory tar after every tenth GPU
178 completion (adapters omitted). Failed directories are isolated, not deleted. A four-task rerun wave
179 (declared before any AUC) then ran 2026-08-25T06:19Z–13:11Z: BIRD dual-sha checker, TyDiQA
180 and SuperNI task419 with a disclosed 1×16 OOM fallback (effective batch still 16), APPS at an
181 8 h ops fence. Analysis was sealed the same day at $n=61$ under the honesty clause. No AUC was
182 computed until that script ran.

183 2.4 Human Interventions

184 Human interventions were discrete, dated, and written into the ledger before the next GPU step. We
185 list the ones that changed what the sealed table contains, including the agent’s own misreports.

186 **False-negative connectivity.** The agent reported that the GPU worker was unreachable. A follow-
187 up over an already-open multiplexed session returned ok. The cause was the agent’s tool layer
188 refusing the remote-shell call, not a host or authentication failure. Production started after that
189 correction, not after a re-provision. That is an agent error in the log, not a cluster outage.

190 **Procedure slip, then a STOP.** Gate 1 (WinoGrande, then Spider) was started before Gate 0’s five-
191 eye table had been accepted. The operator halted Spider at the S4 boundary (`SIGTERM`; S0–S3
192 kept), kept the WinoGrande directory, and only then passed Gate 0. Spider later resumed from that
193 journal boundary and sealed. The STOP did what a comment in chat cannot: it prevented a protocol
194 decision from being made on a moving GPU.

195 **Format overwrite, not EOS.** GSM8K returned $\Delta_{\text{full}} = -0.230$ with unparseable 0/200 and a
196 deterministic greedy rerun. A truncation/EOS reading does not fit those numbers. Base format-
197 compliance was ##### 58 / last-number 142 / none 0; after the probe and after full training the slice
198 was ##### 200/200. The operator’s Gate 0 ruling — format overwrite with reasoning collapse — is
199 the one we keep. The derived `format_compliance` field was backfilled into metrics and did not
200 bump the protocol version.

201 **Schema lag on BIRD, failed closed.** The first BIRD slot died in S0 on `KeyError: sha256`:
202 the checker wanted a single `source.sha256`; the package stored `train_zip_sha256` and
203 `dev_zip_sha256`. No metrics were written. The directory was isolated. The checker was aligned to
204 the dual-sha fields; the bird-only package was not rebuilt. A later slot passed S0, then died of CUDA
205 OOM at the first probe backward pass (S1 pass@1 0.085 / pass@8 0.23; still no `metrics.json`).
206 We did not invent a SQL label to close the queue.

207 **Wall as an ops fence, not a measurement.** APPS hit the 3h cap in S1 with no metrics
208 (over_budget). The 8h rerun was declared before any APPS number existed. It finished S1 at
209 base pass@1 0.0 / pass@8 0.005 and died of CUDA OOM at full-training step 22/495 — still no
210 Δ_{full} . The floor is a missing-negative footnote, not a row.

211 **OOM fallback, two tasks only.** TyDiQA and SuperNI task419 died of CUDA OOM in S4 on
212 the first pass. A run-level 1×16 fallback (effective batch 16, yaml untouched, flag in STATUS and
213 metrics) was declared before those reruns, and only those two task ids may carry the flag. Both
214 resealed: $\Delta_{\text{full}} + 0.530 / +0.505$. ARC-Easy stayed PARTIAL (unparseable 10/200 = 5%, metrics
215 present) and entered the main sample; sensitivity (a) drops it.

216 **Campaign close.** BIRD and APPS stay out: excluded for logistics, before any label
217 existed. The go threshold, the protocol file, and the honesty clause were not reopened after seeing
218 AUC.

219 **What the operator always owns.** The following decisions never moved to the agent, even when
220 the agent drafted the patch: freeze or not freeze the protocol file; keep or replace the 10×100 probe;
221 the honesty clause and the go threshold; whether Mini-Dev may substitute for BIRD; the exact
222 exclude sentence and the rule that it may be used only before a label exists; whether a PARTIAL
223 with metrics (ARC-Easy) enters the sample; whether the pod may be killed; the disclosure boxes in
224 this PDF. The agent owns packing, the unattended queue, harvest SHA checks, five-eye tables, and
225 writing PARTIAL instead of a guessed Δ . When those two columns mixed — the SSH misreport,
226 Gate 1 started early — the log names the mix rather than rewriting it.

227 2.5 Verification

228 Four checks sit under every sealed row, a fifth sits under the paper’s numbers, and a sixth — inde-
229 pendent re-implementation — is the one that actually caught something.

230 Gates (automatic, do not block the queue): base unparseable < 5%; pilot loss must fall; the sys-
231 tems block must be complete (torch, transformers, CUDA, driver, GPU name, base revision, seeds,
232 dry_run=false); a greedy rerun of the unadapted base must match per-id pass (full 200 on list
233 items 1 and every 10th; otherwise 30 rows; seed 20260820). Failure writes STATUS=PARTIAL.
234 Smoke three-task five-eye was all green (GSM8K/WinoGrande unparseable 0/200, Spider 1/200 =
235 0.5%; rerun mismatch 0).

236 Packages carry MANIFEST.sha256. Incremental harvest tars omit adapters and were SHA-
237 checked against the worker; the full archive (adapters and isolated dirs included) has sha256
238 71309f7c...f4028f7d. Figures in Part 1 are redrawn from a frozen plot payload, not from a second
239 analysis.

240 **The check that caught a mis-attribution.** The revision’s headline — “after removing the post-
241 decision feature the gate is significant” — survived code review and survived rerunning the authors’
242 own script. It did not survive an *independent* re-implementation of the LOTO pipeline: that repro-
243 duced AUC 0.663, not 0.844. A cell-by-cell diff of the two 61×10 design matrices found exactly
244 one disagreement, a format-compliance value for MATH, and a decomposition showed that remov-
245 ing leakage moves the AUC *down* while that single cell moves it up by 0.182. The lesson is narrow
246 and reusable: re-running the pipeline verifies determinism, not correctness, and neither a frozen
247 protocol nor a written contract detects a feature-definition change that both the code and its author
248 believe is a bug fix. Only a second implementation compared at the level of the design matrix does.
249 The finding is disclosed in ANALYSIS_PREREG_V2 as a late declaration and the paper’s claim was
250 moved onto a model invariant to it.

251 The analysis contract — go iff $\Delta_{\text{full}} > 0$, task-level LOTO logistic, Spearman on the two deltas,
252 text-only control, bootstrap 95% CI, “AUC non-significant = publish negative” — was committed
253 before the remaining 60 GPU slots. The coding agent drafted this Part 2 from that ledger and those
254 reports. A human author has to walk every claim before the disclosure boxes below can be ticked;
255 they are unchecked in this draft.

256 2.6 Significance of result

257 The qualifying result is a 61-row, execution-based ground-truth table that did not exist before the
258 harness ran, plus a pre-registered reading of it: forging is the default (53/61), the 10-shot probe
259 ranks gain, and the binary gate is not established. Small-model post-training spends real GPU time
260 on tasks that look like GSM8K (negative transfer) and on tasks that look unimportant until a probe
261 is actually run (description-only AUC below chance). A ranker that is honest about not being a
262 stoplight is usable for budget allocation; a non-significant AUC hidden by post-hoc features would
263 not be. The process claim, for this workshop, is narrower: a frozen protocol, a pre-registered honesty
264 clause, and fail-closed logistics were enough to publish that negative gate rather than wait for more
265 no-go tasks.

266 2.7 Interpretability

267 Protocol files, task packages, and run directories are append-only versioned (protocol_v2,
268 tasks_v3 leftover, tasks_v4 59-pack, tasks_v5 bird-only). Old versions remain on disk; a run
269 directory names the protocol and package it consumed. The ledger is append-only. Journal lines are
270 stage-boundary events; Spider is the existence proof that resume does not replay S0–S3. Harvest
271 is two-tier by design: incremental tars for monitoring (no adapters), one full tar for audit. Isolated
272 failed dirs live inside that tar. A sealed run directory is meant to be readable without the agent:
273 task.json and the package manifest, journal.jsonl stage boundaries, eval*_greedy.jsonl
274 for every labelled slice, STATUS, and metrics.json with the systems block (torch / transformers
275 / CUDA / driver / GPU name / pinned base revision / seeds / dry_run=false). If that bundle is
276 missing, the row is not a label. That is why BIRD and APPS are logistics excludes rather than no-go.

277 2.8 Limitations

278 The harness measures one base model, one LoRA recipe, greedy pass@1, and a $\Delta_{\text{full}} > 0$ gate that is
279 empirically easy (8 no-go out of 61). Cluster AUCs were pre-declared and are not estimable. BIRD
280 and APPS produced no label; we do not read that as “this model cannot learn SQL or competition-
281 style code.” Campaign GPU-hours were not summed and are not invented here. We report only
282 walls that sit in sealed reports: smoke-task S0→S6 of GSM8K ≈ 2.16 h, WinoGrande ≈ 0.51 h,
283 Spider ≈ 2.22 h effective (plus a killed half S4 ≈ 0.9 h not counted in the 2.22); a TDP-times-wall
284 upper bound of ≈ 1.5 kWh on that three-task block, not a metered reading; serial production ≈ 39 h
285 wall for 60 slots; Wave-1 TyDiQA ≈ 92 min and task419 ≈ 53 min inside the 3 h cap. The agent’s
286 connectivity misreport and the Gate 1-before-Gate 0 start are in the log because hiding them would
287 make the STOP discipline look cleaner than it was.

3 Broader Impact

Guardrails earned their keep when they failed *closed*. The BIRD checker aborted on a missing sha256 key rather than packing a substitute hash or writing a fake `metrics.json`; the later CUDA OOM likewise produced no label. GSM8K’s negative Δ_{full} came with unparseable 0/200 and a format-compliance shift from mixed `####/last-number` at base to `#### 200/200` after training — evidence against an EOS/truncation story, and the reason the operator could rule “format overwrite” instead of discarding a negative as a generation glitch. A rail that invents a number to keep a queue green is not a rail.

Pre-registration is what made the negative primary endpoint publishable. The honesty clause (bootstrap CI covering 0.5 equals a negative, to be printed) was committed before the remaining sixty GPU slots and before any AUC existed. After the seal, the tempting moves — add a feature, drop GSM8K, move the go threshold, reopen BIRD — were exactly the moves the contract forbids. With 8 no-go tasks the gate was always going to be under-powered; the clause prevented that fact from being negotiated away in the write-up.

STOP points were the human-agent interface, with a mixed record. The protocol-v3 STOP (keep 10×100 ; recitation is the probe) and the analysis STOP (no production before the SuperNI ids and the pre-reg were on the mainline) are the reason Part 1 has one recipe and one contract. The cost is real: Gate 1 started before Gate 0 was accepted, and the connectivity misreport looked like downtime until a human asked for a one-line multiplex check. A STOP that exists only as a chatbot apology does not stop a GPU.

Diagnosis was split on purpose. The agent is responsible for five-eye tables, SHA checks, journal resume, and for writing PARTIAL rather than a guessed Δ . The operator is responsible for mechanism rulings (format overwrite), for ops-fence vs measurement (APPS 3 h then 8 h), and for the exclude sentence that may be used only *before any label existed*. We would like the field to copy that split: freeze the recipe, pre-register the honesty clause, fail closed, and keep the operator on the one-way door out of the sample.

4 Disclosure Statement

Agent(s) / model(s) used The 61-row experimental table was produced by a frozen Python harness on a single serial GPU queue (protocol v2; base model Qwen2.5-1.5B-Instruct at the pinned revision in the protocol file). An LLM coding agent packed tasks, monitored the queue, diagnosed failures, executed the pre-registered analysis and the declared CPU-only revision, and drafted Parts 1–3 of this manuscript, including figures generated from the frozen plot payload. The revision’s own attribution error was found by an independent re-implementation of the analysis and is disclosed as a late declaration in the revision contract; no experimental number was recomputed on new hardware. The coding agent’s model identity is omitted here to preserve double-blind review and will be named to the program chairs. No experimental number in Part 1 was computed outside the sealed analysis script. Human status of this draft: data, hashes, and analysis artifacts were produced under the harness and the pre-registration; this text has *not* yet been line-checked by a human author. The two attestations below are therefore left unchecked.

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